

PROOFS OF IRRATIONALITY

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ABSTRACT. We start by looking at some basic properties regarding operations with rational and irrational numbers. We then go on to show that certain radicals are irrational. Next, we state and prove a criterion for irrationality and use it to prove that e is irrational. After this, we look at alternate proofs of this fact, and show that e^r is irrational for all nonzero rationals r . Then, we see that π and π^2 are irrational, before going on to prove the same for $\cos(1)$. We look at other irrational values related to the trigonometric functions next, and prove that $\ln(r)$ is irrational for positive rational $r \neq 1$. We follow this up by looking at other logarithms, and proving that $\cosh(1)$ is irrational. We then discuss transcendence, and use the Lindemann–Weierstrass theorem to reprove many of our earlier statements. We end with posing open problems in irrationality.

1. CLOSURE OF IRRATIONAL NUMBERS

In this section, we shall look at some basic theorems regarding operations with rational and irrational numbers. All numbers in this paper will be assumed to be real unless they are defined during our discussion of transcendence. We will also assume all denominators to be nonzero.

Theorem 1.1. *If $r \in \mathbb{Q}$, and $a \notin \mathbb{Q}$, then the following are true.*

- (1) $r + a \notin \mathbb{Q}$.
- (2) $ra \notin \mathbb{Q}$.
- (3) $-a \notin \mathbb{Q}$.
- (4) $a^{-1} \notin \mathbb{Q}$.

Proof. Our methods for all of these will be proof by contradiction.

- (1) If we assume that $r + a \in \mathbb{Q}$, we will let it be $r_0 \in \mathbb{Q}$. Then, $a = r_0 - r$. Because $\mathbb{Q}$ is additively closed, this means that $a \in \mathbb{Q}$, a contradiction.
- (2) If we assume that $ra \in \mathbb{Q}$, we will let it be $r_0 \in \mathbb{Q}$. Then, $a = \frac{r_0}{r}$. Because $\mathbb{Q}$ is multiplicatively closed, this means that $a \in \mathbb{Q}$, a contradiction.
- (3) If we assume that $-a \in \mathbb{Q}$, we will let it be $\frac{p}{q} \in \mathbb{Q}$. Then, $a = -\frac{p}{q}$, meaning that it is rational, which is a contradiction.
- (4) If we assume that $a^{-1} \in \mathbb{Q}$, we will let it be $\frac{p}{q} \in \mathbb{Q}$. Then, $a = \frac{q}{p}$, meaning that it is rational, which is a contradiction.

■

2. IRRATIONAL RADICALS

2.1. Square Roots of Primes. It is well known that $\sqrt{2}$ is irrational. We will present the classic proof here.

Proposition 2.1. $\sqrt{2}$ is irrational.

Proof. Let us assume that $\sqrt{2} = \frac{a}{b}$, where a and b are relatively prime integers. Then, $2 = \frac{a^2}{b^2}$, or $2b^2 = a^2$. Now, if a was odd, then $a^2 = (2k+1)^2 = 4k^2 + 4k + 1$ is also odd, so a must be even. We will write $a = 2k$. Now, we have that $2b^2 = (2k)^2 = 4k^2$. Dividing by 2 gives $b^2 = 2k^2$. Therefore, b must be even. However, this contradicts our assumption that a and b are relatively prime. Therefore, $\sqrt{2}$ is irrational. ■

We can also show this for $\sqrt{3}$.

Proposition 2.2. $\sqrt{3}$ is irrational.

Proof. Let us suppose that $\sqrt{3} = \frac{a}{b}$ where a and b are relatively prime integers. We can say that $3b^2 = a^2$. Now, if we assume that $3 \nmid a$, then (as $3 \mid a^2$), $3 \mid a$, which is a contradiction. Therefore, we can write $a = 3k$, and get $3b^2 = (3k)^2 = 9k^2$. We can divide by 3 to get $b^2 = 3k^2$, so we can write $b = 3l$. This contradicts our assumption that a and b are relatively prime, so $\sqrt{3}$ is irrational. ■

We can generalize this proofs easily to any prime. We shall first prove a lemma.

Lemma 2.3. If $p \mid a^2$ for a prime p , then $p \mid a$.

Proof. Let us suppose that $p \nmid a$. Because $p \mid a \cdot a$, and $p \nmid a$, we have that $p \mid a$. This is a contradiction, and so $p \mid a$. ■

Theorem 2.4. If p is a prime, then $\sqrt{p}$ is irrational.

Proof. Let us suppose that $\sqrt{p} = \frac{a}{b}$ where a and b are relatively prime integers. Then, $pb^2 = a^2$. We write $a = pk$ to get $pb^2 = (pk)^2 = p^2k^2$. Dividing by p gives $b^2 = pk^2$, and so we can write $b = pl$, meaning that a and b are not relatively prime, which is a contradiction. Therefore, $\sqrt{p} \notin \mathbb{Q}$. ■

2.2. More Interesting Square Roots. We want to show that other roots are irrational. We will show this for $\sqrt{6}$ below.

Proposition 2.5. $\sqrt{6}$ is irrational.

Proof. Let us suppose that $\sqrt{6} = \frac{a}{b}$ where a and b are relatively prime integers. Then, $6b^2 = a^2$. We can write $a = 2k$ to get $6b^2 = 4k^2$. Thus, $3b^2 = 2k^2$, so we can write $b = 2l$. This contradicts our assumption that $\gcd(a, b) = 1$, so $\sqrt{6} \notin \mathbb{Q}$. ■

Note that we could have said $a = 3k$, and used this to show that $\sqrt{6}$ is irrational. We can generalize the proof of this.

Theorem 2.6. If p and q are distinct primes, then $\sqrt{pq}$ is irrational.

Proof. Let us suppose that $\sqrt{pq} = \frac{a}{b}$ where a and b are integers with $\gcd(a, b) = 1$. Then, $pqb^2 = a^2$, so we can write $a = pk$ to get $pqb^2 = p^2k^2$. Therefore, $qb^2 = pk^2$, and (as $q \neq p$ is prime) we can say $b = pl$. This means that $\gcd(a, b) \neq 1$, which is a contradiction. ■

2.3. Sums of Square Roots. We want to show that sums of square roots are also irrational. We will prove this for an example first.

Proposition 2.7. $\sqrt{2} + \sqrt{3}$ is irrational.

Proof. Let us suppose that $\sqrt{2} + \sqrt{3} = \frac{a}{b}$ for integers a and b such that $\frac{a}{b}$ is fully reduced. Then, $(\sqrt{2} + \sqrt{3})^2 = \frac{a^2}{b^2}$. This is equal to $\sqrt{2}^2 + 2\sqrt{2}\sqrt{3} + \sqrt{3}^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}$. Thus, $\sqrt{6} = \frac{a^2 - 5b^2}{2b^2}$, which means that $\sqrt{6} \in \mathbb{Q}$. This is a contradiction, so $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$. ■

We can now see how to easily generalize this proof.

Theorem 2.8. If p and q are distinct primes, then $\sqrt{p} + \sqrt{q}$ is irrational.

Proof. Let us suppose that $\sqrt{p} + \sqrt{q} = \frac{a}{b}$, where we have completely reduced fraction $\frac{a}{b}$ with $a, b \in \mathbb{Z}$. Then, $(\sqrt{p} + \sqrt{q})^2 = p + q + 2\sqrt{pq} = \frac{a^2}{b^2}$. However, this implies that $\sqrt{pq}$ is rational, which is a contradiction. Therefore, $\sqrt{p} + \sqrt{q}$ is irrational. ■

2.4. More General Roots. We will show that $2^{1/n}$ is irrational for integers $n \geq 2$.

Theorem 2.9. $2^{1/n}$ is irrational for $n \in \mathbb{Z} \geq 2$.

Proof. If we assume that $2^{1/n} = \frac{a}{b}$ where $a, b \in \mathbb{Z}$ are relatively prime, then we can say $2b^n = a^n$. Thus, we can write $a = 2k$ and derive $b^n = 2^{n-1}k^n$. This means that b is also even, creating a contradiction. ■

We can in fact prove that $2^{a/b}$ is irrational for positive integers a and b such that $\gcd(a, b) = 1$, with $a \neq 0$ and $b \neq 1$. To do this, we shall use the rational root theorem.

Lemma 2.10 (Rational Root Theorem). If a fully reduced fraction $\frac{p}{q}$ is a root of $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, then $p|a_0$, and $q|a_n$.

Proof. Let us suppose that $\frac{p}{q}$ is a root. Then, we can say $a_n \frac{p^n}{q^n} + a_{n-1} \frac{p^{n-1}}{q^{n-1}} + \dots + a_1 \frac{p}{q} + a_0 = 0$. Multiplying by q^n gives $a_n p^n + a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1} + a_0 q^n = 0$. We can rearrange this to get $a_n p^n + a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1} = -a_0 q^n$. Note that the left hand side is divisible by p . Because $\gcd(p, q) = 1$, this means that $p|a_0$. Similarly, we have that $a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1} + a_0 q^n = -a_n p^n$. Because q divides the left hand side, it must divide a_n . ■

Theorem 2.11. For a fully reduced fraction $\frac{a}{b}$, where $b \neq 1$ and $a \neq 0$, $2^{a/b}$ is irrational.

Proof. We only need to prove this for $\frac{a}{b} > 0$, because the reciprocal will also be irrational. Therefore, we will assume $\frac{a}{b} > 0$. We can see that $2^{a/b}$ is a root of $x^b - 2^a$. Therefore, the rational roots are of the form $\pm 2^k$ for $k \leq a$. Note that, if b is odd, the roots cannot be negative, and if b is even, then if x is a root, so is $-x$, so we only need to check the positive cases. If this is a root, then $(2^k)^b - 2^a = 0$. This means that $a = kb$, or that $\frac{a}{b} = k$, contradicting the assumption that $\frac{a}{b}$ is a fully reduced fraction with $b \neq 1$. ■

3. AN IRRATIONALITY CRITERION

Here we will prove the following theorem.

Theorem 3.1. A real number α is irrational if and only if, for any $\varepsilon > 0$, there exist integers p and q such that

$$0 < |q\alpha - p| < \varepsilon.$$

3.1. The Dirichlet Approximation Theorem. To prove our desired criterion, we need to use the Dirichlet Approximation Theorem. We will state and prove it here.

Theorem 3.2 (Dirichlet Approximation Theorem). *Let N be a positive integer, and let α be any real number. Then, there exist integers p and q with $1 \leq q \leq N$ such that*

$$|q\alpha - p| < \frac{1}{N}.$$

Proof. We will consider the set $A = \{a_0, a_1, \dots, a_N\}$ where we define $a_i = i\alpha - \lfloor i\alpha \rfloor$. Clearly, $A \subset [0, 1)$. Because $|A| = N + 1$, there is some positive $k \leq N$ so that $[\frac{k-1}{N}, \frac{k}{N})$ contains two elements of A . We will call these two elements a_i and a_j where $i > j$. Thus, $|a_i - a_j| < \frac{1}{N}$. Using our definition of a_i , this means that $|i\alpha - \lfloor i\alpha \rfloor - j\alpha + \lfloor j\alpha \rfloor| < \frac{1}{N}$. We can rearrange this to get $|(i - j)\alpha - (\lfloor i\alpha \rfloor - \lfloor j\alpha \rfloor)| < \frac{1}{N}$. Thus, we can choose $q = i - j$, and $p = \lfloor i\alpha \rfloor - \lfloor j\alpha \rfloor$ to prove the theorem. ■

3.2. The Irrationality Criterion. Now that we have proven the Dirichlet Approximation Theorem, we are ready to prove the irrationality criterion.

Proof of Theorem 3.1. Let us suppose that $\alpha \notin \mathbb{Q}$, and let $\varepsilon > 0$. We shall choose N such that $\frac{1}{N} < \varepsilon$. By the Dirichlet Approximation Theorem, there are integers p and q with $1 \leq q \leq N$ such that $|q\alpha - p| < \frac{1}{N} < \varepsilon$. Because α is irrational, this must be nonzero. Let us now assume that $\alpha \in \mathbb{Q}$. We will write $\alpha = \frac{r}{s}$ for relatively prime integers r and s . Now, if we have $\frac{p}{q} \neq \frac{r}{s}$, where $p, q \in \mathbb{Z}$, then

$$\left| \frac{r}{s} - \frac{p}{q} \right| = \left| \frac{qr - ps}{qs} \right| \geq \frac{1}{qs}.$$

Multiplying by q gives

$$|q\alpha - p| \geq \frac{1}{s}.$$

Thus, if we choose $\frac{1}{\varepsilon} > s$, we get $|q\alpha - p| \geq \frac{1}{s} > \varepsilon$. The only case we have disregarded is if $\frac{p}{q} = \frac{r}{s}$. In this case, $|q\alpha - p| = 0$. Thus, if α is rational, there are not integers p and q such that $0 < |q\alpha - p| < \varepsilon$ for all $\varepsilon > 0$. ■

3.3. Applications of the Irrationality Criterion. We shall use the irrationality criterion to show that e is irrational.

Theorem 3.3. *e is irrational.*

Proof. We know that $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. We will let $N \in \mathbb{Z}_{>0}$ such that $\frac{1}{N} < \varepsilon$. We will let $q = N!$, and p be $\sum_{n=0}^N \frac{N!}{n!}$. Then,

$$\begin{aligned} |qe - p| &= \left| N! \sum_{n=0}^{\infty} \frac{1}{n!} - N! \sum_{n=0}^N \frac{1}{n!} \right| \\ &= N! \sum_{n=N+1}^{\infty} \frac{1}{n!} \\ &= \sum_{n=N+1}^{\infty} \frac{1}{(N+1)(N+2) \cdots (n-1)(n)} \end{aligned}$$

$$\leq \sum_{n=N+1}^{\infty} \frac{1}{(N+1)^{n-N}}.$$

Now, we may use the fact that this is a geometric series to get that this is equal to

$$\frac{1}{N+1} \cdot \frac{1}{1 - 1/(N+1)}.$$

This is equal to

$$\begin{aligned} \frac{1}{N+1} \cdot \frac{N+1}{N} \\ = \frac{1}{N} \\ < \varepsilon. \end{aligned}$$

We can see that $|qe - p| \neq 0$, so, by Theorem 3.1, e is irrational. ■

Theorem 3.1 can also be used to show that $\zeta(3)$ is irrational. This proof can be found in [RSb, Chapter 4].

4. e AND ITS POWERS

4.1. Alternate Proofs of e 's Irrationality. We have already shown that e is irrational using the irrationality criterion. We will present two alternate proofs here.

Second Proof of Theorem 3.3. We know that $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. If we assume that $e = \frac{a}{b}$ for relatively prime integers a and b , then we can say that $n!be = n!a$ for every $n > 0$. We can see that our right hand side is an integer. However, our left side is equal to

$$b \sum_{k=0}^n \frac{n!}{k!} + b \sum_{k=n+1}^{\infty} \frac{n!}{k!}.$$

We can see, however, that the second sum is equal to

$$\frac{b}{(n+1)} + \frac{b}{(n+1)(n+2)} + \cdots.$$

We can see that this sum is greater than $\frac{b}{n+1}$. We can also see that, by comparison with geometric series, that this is less than $\frac{b}{n}$. Thus, for large n , this is not an integer, which is a contradiction. ■

Third Proof of Theorem 3.3. Let us assume that $e = \frac{p}{q}$ where p and q are relatively prime integers. Then, we can write

$$\frac{p}{q} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{q!} + \cdots.$$

Now, we will rearrange this to get

$$\frac{p}{q} - \frac{1}{0!} - \frac{1}{1!} - \cdots - \frac{1}{q!} = \frac{1}{(q+1)!} + \frac{1}{(q+2)!} + \cdots.$$

We can multiply by $q!$ to get

$$\frac{pq!}{q} - \frac{q!}{0!} - \frac{q!}{1!} - \cdots - \frac{q!}{q!} = \frac{q!}{(q+1)!} + \frac{q!}{(q+2)!} + \cdots.$$

It is easy to see that the left hand side is an integer. We can also see, by comparison with geometric series, that our right hand side is less than $\frac{1}{q}$. It is also clearly positive. Thus, this means that there is an integer n such that $0 < n < \frac{1}{q}$, which is a contradiction. ■

Remark 4.1. It is not hard to see that these proofs are very similar. One can in fact modify one proof to obtain the other.

4.2. The Irrationality of e^2 . To show that e^2 is irrational, we will use a similar technique that we used for e .

Theorem 4.2. e^2 is irrational.

Proof. If we assume that $e^2 = \frac{a}{b}$ where $a, b \in \mathbb{Z}$ and $\gcd(a, b) = 1$, then $be = ae^{-1}$. We can use the series $e = \sum_{k=0}^{\infty} \frac{1}{k!}$ and $e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$, and multiply by $n!$. We can see that $n!be$ has an integer part, and a leftover term of

$$n!b \left(\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots \right).$$

We know that this term is between $\frac{b}{n+1}$ and $\frac{b}{n}$. Similarly, $n!ae^{-1}$ has a leftover part of

$$(-1)^{n+1}n!a \left(\frac{1}{(n+1)!} - \frac{1}{(n+2)!} \pm \cdots \right).$$

Now, if n is even, then this is larger than $-\frac{a}{n+1}$, but smaller than

$$-\frac{a}{n+1} + \frac{a}{(n+1)^2} \mp \cdots = -\frac{a}{n+1} \left(\frac{1}{1 + 1/(n+1)} \right) = -\frac{a}{n+1} \left(1 - \frac{1}{n+2} \right) < 0.$$

However, this means that, for large n , $n!ae^{-1}$ is slightly smaller than an integer, while $n!be$ is slightly larger than an integer. This is a contradiction, and so e^2 is irrational. ■

This idea can also be used to show that e^4 is irrational, see [AZHE10, Chapter 8].

4.3. Integer Powers of e . We shall prove the following theorem.

Theorem 4.3. For integers $s \neq 0$, e^s is irrational.

To do this, we shall need to make use of a lemma.

Lemma 4.4. For some $n \geq 1$, let

$$f_n(x) = \frac{x^n(1-x)^n}{n!}.$$

Then, the following are true.

- (1) $f_n(x)$ can be written as $\frac{1}{n!} \sum_{k=n}^{2n} c_k x^k$ where $c_k \in \mathbb{Z}$.
- (2) For $0 < x < 1$, $0 < f_n(x) < \frac{1}{n!}$.
- (3) For all $k > 0$, $f_n^{(k)}(0)$ and $f_n^{(k)}(1)$ are integers.

Proof. Part (1) is clear, and part (2) can be checked by basic differentiation, so we will prove only part (3). Note that, $f_n^{(k)}(0)$ is 0 unless $n \leq k \leq 2n$. Because we are plugging in 0, we only need to worry about the constant term. The constant term for these are $\frac{k!}{n!}c_k$, which is an integer in the range. Now, notice that $f_n(x) = f_n(1-x)$, so $f_n^{(k)}(x) = (-1)^k f_n^{(k)}(1-x)$, which means that $f_n^{(k)}(1) = (-1)^k f_n^{(k)}(0)$, which is an integer. ■

We are now able to prove our theorem.

Proof of Theorem 4.3. Let us suppose that $e^s = \frac{a}{b}$ for relatively prime integers a and b . We will let n be determined later, and we will let

$$F(x) := s^{2n}f(x) - s^{2n-1}f'(x) + s^{2n-2}f''(x) \mp \cdots + f^{(2n)}(x),$$

where we write $f(x)$ for $f_n(x)$ from the above lemma. Because $f^{(k)}(x) = 0$ for $k > 2n$, we can write

$$F(x) = s^{2n}f(x) - s^{2n-1}f'(x) + s^{2n-2}f''(x) \mp \cdots.$$

This allows us to say

$$F'(x) = -sF(x) + s^{2n+1}f(x).$$

Now, let us look at $\frac{d}{dx}(e^{sx}F(x))$. The product rule gives

$$\begin{aligned} \frac{d}{dx}(e^{sx}F(x)) &= se^{sx}F(x) + e^{sx}F'(x) = se^{sx}F(x) + e^{sx}(-sF(x) + s^{2n+1}f(x)) \\ &= s^{2n+1}e^{sx}f(x). \end{aligned}$$

This allows us to find

$$N := b \int_0^1 s^{2n+1}e^{sx}f(x)dx = b(e^{sx}F(x))|_0^1 = be^sF(1) - bF(0) = aF(1) - bF(0).$$

Part (3) of our lemma implies that this is an integer. Part (2) of the lemma gives us bounds, which allow us to say that

$$0 < N < \frac{bs^{2n}}{n!} \int_0^1 se^{sx}dx = \frac{bs^{2n}}{n!}(e^s - 1).$$

We can assume that s is positive, and the closure properties will cover the case of $s < 0$. We will also let n be such that $n! > as^{2n+1}$. Thus,

$$N < \frac{bs^{2n+1}e^s}{n!} = \frac{as^{2n+1}}{n!} < 1.$$

This is a contradiction, and so e^s is irrational. ■

4.4. Rational Powers of e . We have shown that for $n \in \mathbb{Z} \setminus \{0\}$, e^n is irrational. For $n \in \mathbb{R} \setminus \{0\}$, this is clearly false (for example, $n = \ln(2)$). We wish to determine this for the rationals.

Theorem 4.5. *If r is a nonzero rational number, then e^r is irrational.*

Proof. We will write $r = \frac{p}{q}$ for $p, q \in \mathbb{Z}$ and $\gcd(p, q) = 1$. Let us suppose that $e^{p/q} = \frac{a}{b}$ for relatively prime integers a and b . This means that $e^p = \frac{a^q}{b^q}$. This means that e^p is rational, a contradiction. ■

5. PI, TRIGONOMETRIC, LOGARITHMIC, AND HYPERBOLIC VALUES

5.1. π and π^2 's Irrationality. We will first prove the well-known theorem that π is irrational.

Theorem 5.1. *π is irrational.*

Proof. We will let $\pi = \frac{a}{b}$ where a and b are integers with $\gcd(a, b) = 1$. We define

$$f(x) := \frac{x^n(a - bx)^n}{n!}.$$

Note that we can make an analog of Lemma 4.4 into a lemma based around $f(x)$. Proving the exact details of this analog is left as an exercise for the reader. We will now define

$$F(x) = f(x) - f^{(2)}(x) + f^{(4)}(x) \mp \cdots.$$

Notice that we can have this infinite series because the derivatives beyond $2n$ vanish. We also have that $F''(x) = -F(x) + f(x)$. Basic calculus gives that

$$\begin{aligned} \frac{d}{dx} (F'(x) \sin(x) - F(x) \cos(x)) &= F''(x) \sin(x) + F'(x) \cos(x) - F'(x) \cos(x) + F(x) \sin(x) \\ &= F''(x) \sin(x) + F(x) \sin(x) = (-F(x) + f(x)) \sin(x) + F(x) \sin(x) = f(x) \sin(x). \end{aligned}$$

Thus,

$$\int_0^\pi f(x) \sin(x) dx = (F'(x) \sin(x) - F(x) \cos(x)) \Big|_0^\pi = F(\pi) + F(0).$$

This is an integer. However, we can also establish a bound on this for $0 < x < \pi$:

$$0 < f(x) \sin(x) < f(x) < \frac{\pi^n a^n}{2^{2n} n!}.$$

Thus, our integral (which is an integer) is arbitrarily small for large enough n , and so we have a contradiction. Thus, π is irrational. ■

Corollary 5.2. *For positive integers n , $\pi^{1/n}$ is irrational.*

Proof. Let us suppose that $\pi^{1/n}$ is rational, say $\frac{a}{b}$. Then, we have that $\frac{a^n}{b^n} = \pi$. This is a contradiction, and so $\pi^{1/n}$ is irrational. ■

Now, we will do the same for π^2 . When we write $f(x)$, we will mean $f_n(x)$ from Lemma 4.4.

Theorem 5.3. *π^2 is irrational.*

Proof. Let us assume $\pi^2 = \frac{a}{b}$ where $a, b \in \mathbb{Z}$, and are relatively prime. We define

$$F(x) := b^n (\pi^{2n} f(x) - \pi^{2n-2} f^{(2)}(x) + \pi^{2n-4} f^{(4)}(x) \mp \cdots).$$

Note that this satisfies $F''(x) = -\pi^2 F(x) + b^n \pi^{2n+2} f(x)$. We can get that

$$\begin{aligned} \frac{d}{dx} (F'(x) \sin(\pi x) - \pi F(x) \cos(\pi x)) &= F''(x) \sin(\pi x) + \pi^2 F(x) \sin(\pi x) \\ &= b^n \pi^{2n+2} f(x) \sin(\pi x) = \pi^2 a^n f(x) \sin(\pi x). \end{aligned}$$

Therefore,

$$N := \pi \int_0^1 a^n f(x) \sin(\pi x) dx = (\pi^{-1} F'(x) \sin(\pi x) - F(x) \cos(\pi x)) \Big|_0^1 = F(1) + F(0).$$

We can see that $N \in \mathbb{Z}$. Now, let us choose n so large such that $\frac{2a^n}{n!} < 1$. Our bounds on $f(x)$ allow us to say that

$$0 < N = \pi \int_0^1 a^n f(x) \sin(\pi x) dx < \pi a^n \frac{1}{n!} \int_0^1 \sin(\pi x) dx = \frac{2a^n}{n!} < 1.$$

This is a contradiction. ■

5.2. The Irrationality of $\cos(1)$. Here we will prove that $\cos(1)$ is irrational using the irrationality criterion.

Proposition 5.4. $\cos(1)$ is irrational.

Proof. We know that $\cos(1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}$. We will choose N such that $\frac{1}{N} < \varepsilon$. We will let $q = N!$ and $p = N! \sum_{n=0}^N \frac{(-1)^n}{(2n)!}$. Then, we have that

$$\begin{aligned} |q \cos(1) - p| &= \left| N! \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} - N! \sum_{n=0}^N \frac{(-1)^n}{(2n)!} \right| \\ &= \sum_{n=N+1}^{\infty} \frac{(-1)^n N!}{(2n)!} \leq \sum_{n=N+1}^{\infty} \frac{(-1)^n}{(N+1)^{2n-N}} \leq \sum_{n=N+1}^{\infty} \frac{1}{(N+1)^{2n-N}} \\ &= \frac{1}{(N+1)^{N+2}} \cdot \frac{1}{1 - 1/(N+1)^2} = \frac{1}{(N+1)^{N+2}} \cdot \frac{(N+1)^2}{(N+1)^2 - 1} \\ &= \frac{1}{N} \cdot \frac{1}{(N+1)^N} \cdot \frac{1}{N+2} < \frac{1}{N} < \varepsilon. \end{aligned}$$

Therefore, by Theorem 3.1, $\cos(1)$ is irrational. ■

This proof can be easily modified to show that $\sin(1)$ is irrational, and is left as an exercise to the reader.

5.3. Other Trigonometric Values. Many values of trigonometric functions, such as $\cos\left(\frac{\pi}{9}\right)$ and $\sin\left(\frac{\pi}{18}\right)$ are irrational. We will show these here.

Proposition 5.5. $\cos\left(\frac{\pi}{9}\right)$ is irrational.

Proof. By the triple angle formula, we have that $\cos\left(\frac{\pi}{3}\right) = 4\cos^3\left(\frac{\pi}{9}\right) - 3\cos\left(\frac{\pi}{9}\right)$. Writing x for $\cos\left(\frac{\pi}{9}\right)$ gives $\frac{1}{2} = 4x^3 - 3x$, or $8x^3 - 6x - 1 = 0$. The rational roots of these must be $\pm 1, \pm \frac{1}{2}, \pm \frac{1}{4},$ or $\pm \frac{1}{8}$. Direct substitution gives that none of these are actual roots. Thus, as $\cos\left(\frac{\pi}{9}\right)$ is a root of this, $\cos\left(\frac{\pi}{9}\right)$ is irrational. ■

We can prove that $\sin\left(\frac{\pi}{18}\right)$ is irrational 2 different ways.

Proposition 5.6. $\sin\left(\frac{\pi}{18}\right)$ is irrational.

First Proof of Proposition 5.6. From the triple angle formula for $\sin$, we have that $\sin\left(\frac{\pi}{6}\right) = 3\sin\left(\frac{\pi}{18}\right) - 4\sin^3\left(\frac{\pi}{18}\right)$, meaning that $\sin\left(\frac{\pi}{18}\right)$ is a root of $\frac{1}{2} = 3x - 4x^3$, or $8x^3 - 6x + 1 = 0$. We can once again use the rational root theorem to show that this has no rational roots, and so $\sin\left(\frac{\pi}{18}\right)$ is irrational. ■

Second Proof of Proposition 5.6. We have that, from the cosine double angle formula, $\cos\left(\frac{\pi}{9}\right) = 1 - 2\sin^2\left(\frac{\pi}{18}\right)$. Thus, if $\sin\left(\frac{\pi}{18}\right)$ were rational, then $\cos\left(\frac{\pi}{9}\right)$ would be too. This is a contradiction, so $\sin\left(\frac{\pi}{18}\right)$ must be irrational. ■

5.4. Inverse Trigonometric Functions. We will prove a result regarding the irrationality of inverse trigonometric functions.

Theorem 5.7. *For every odd integer $n \geq 3$,*

$$\frac{1}{\pi} \cos^{-1} \left(\frac{1}{\sqrt{n}} \right)$$

is irrational.

Recall that, letting $n = 9$, this is useful in the solution of Hilbert's Third Problem. In the proof, we shall make use of a lemma.

Lemma 5.8. $\cos((k+1)\varphi) = 2 \cos(k\varphi) \cos(\varphi) - \cos((k-1)\varphi)$.

Proof. Recall that $\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta)$, and that $\cos(\alpha - \beta) = \cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta)$. Adding these gives that

$$\begin{aligned} \cos(\alpha + \beta) + \cos(\alpha - \beta) &= \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) + \cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta) \\ &= 2 \cos(\alpha) \cos(\beta). \end{aligned}$$

If we substitute $\theta = \alpha + \beta$ and $\phi = \alpha - \beta$, then $\alpha = \frac{\theta + \phi}{2}$, and $\beta = \frac{\theta - \phi}{2}$. This gives the formula

$$\cos(\theta) + \cos(\phi) = 2 \cos \left(\frac{\theta + \phi}{2} \right) \cos \left(\frac{\theta - \phi}{2} \right).$$

Now, if we let $\theta = (k+1)\varphi$, and $\phi = (k-1)\varphi$, we get that

$$\cos((k+1)\varphi) + \cos((k-1)\varphi) = 2 \cos(k\varphi) \cos(\varphi),$$

which is the same thing as saying that

$$\cos((k+1)\varphi) = 2 \cos(k\varphi) \cos(\varphi) - \cos((k-1)\varphi).$$

■

Proof of Theorem 5.7. We shall define $\varphi_n = \cos^{-1} \left(\frac{1}{\sqrt{n}} \right)$. We wish to show that

$$\cos(k\varphi_n) = \frac{A_k}{\sqrt{n}^k},$$

where A_k is an integer such that $n \nmid A_k$. This can be done with induction on k . For $k = 0, 1$, this is true with $A_k = 1$. We can use our formula to show this for $k + 1$:

$$\cos((k+1)\varphi_n) = 2 \frac{1}{\sqrt{n}} \frac{A_k}{\sqrt{n}^k} - \frac{A_{k-1}}{\sqrt{n}^{k-1}} = \frac{2A_k - nA_{k-1}}{\sqrt{n}^{k+1}} := \frac{A_{k+1}}{\sqrt{n}^{k+1}}.$$

It is clear that, if $n \nmid A_k$, then $n \nmid A_{k+1}$. Now, let us assume that $\frac{1}{\pi} \varphi_n = \frac{k}{l}$ for relatively prime integers k and l . Then, we have that $l\varphi_n = k\pi$. Taking the cosine of both sides gives $\frac{A_l}{\sqrt{n}^l} = \cos(l\varphi_n) = \cos(k\pi) = \pm 1$. Therefore, $\sqrt{n}^l = \pm A_l$. It is clear that $l \neq 0$, and that this doesn't work for $l = 1$. Therefore, we can assume that $l \geq 2$. In this case, $n \mid \sqrt{n}^l$. From the fact that $\sqrt{n}^l \mid A_l$, we have that $n \mid A_l$, which is a contradiction. ■

5.5. Irrationality of Natural Logarithms. It is well known that $\ln(2)$ is irrational. We shall prove this here.

Proposition 5.9. $\ln(2)$ is irrational.

Proof. Let us assume that $\ln(2) = \frac{a}{b}$ with $a, b \in \mathbb{Z}$ and $\gcd(a, b) = 1$. Then, $2 = e^{a/b}$. However, we know that $e^{a/b}$ must be irrational, so this is a contradiction. ■

It is clear that there is nothing about 2 in this situation, and so we can prove this for (almost) any rational number.

Theorem 5.10. If $r \in \mathbb{Q}_{>0} \setminus \{1\}$, then $\ln(r)$ is irrational.

Proof. If we assume that $\ln(r) = \frac{a}{b}$ where a and b are integers with $\gcd(a, b) = 1$, then we can say that $r = e^{a/b}$. However, this means that r must be irrational, which is a contradiction. ■

5.6. Irrationality of Logarithms with Integer Bases. A well known irrational number is $\log_2(3)$. We shall prove that this is irrational.

Proposition 5.11. $\log_2(3)$ is irrational.

Proof. Let us suppose that $\log_2(3) = \frac{a}{b}$ with $a, b \in \mathbb{Z}$ and $\gcd(a, b) = 1$. Then, $3 = 2^{a/b}$, or $3^b = 2^a$. By the Fundamental Theorem of Arithmetic, this is only possible if $a = b = 0$. Therefore, $\log_2(3)$ is irrational. ■

It is clear that there is nothing special about 3 here, so we will prove this generally.

Proposition 5.12. If $n \in \mathbb{Z}$ is not a power of 2, then $\log_2(n)$ is irrational.

Proof. Let us write $n = 2^p a$ such that a is odd. Now, let us assume that $\log_2(n) = \frac{b}{c}$ where b and c are relatively prime positive integers. Then, $2^b = 2^{pc} a^c$. This is only possible if either $a = 1$, or $c = 0$, which contradict our initial conditions. Therefore, $\log_2(n)$ is irrational. ■

We will show that the same is true for 10.

Proposition 5.13. If $n \in \mathbb{Z}$ is not a power of 10, then $\log_{10}(n)$ is irrational.

Proof. Let us write $n = 2^{p_1} 5^{p_2} a$ such that a is relatively prime to 10. If we assume that $\log_{10}(n) = \frac{b}{c}$ where b and c are relatively prime positive integers, then we have that $10^b = 2^{p_1 c} 5^{p_2 c} a^c$. If $a \neq 1$, then this is impossible. If $a = 1$, then we have that $p_1 c = p_2 c = b$. However, this means that $p_1 = p_2$, or that n is a power of 10, which is a contradiction. ■

This may lead us to believe that this is true for every integer. However, this is false.

Counterexample. $\log_8(16) = \frac{4}{3}$.

5.7. Irrationality of $\cosh(1)$. Recall that $\cosh(x)$ is defined as

$$\cosh(x) = \frac{e^x + e^{-x}}{2},$$

and that $\sinh(x)$ is defined as

$$\sinh(x) = \frac{e^x - e^{-x}}{2}.$$

Remark 5.14. If the reader is familiar with the complex definitions of $\sin(x)$ and $\cos(x)$, they may notice that $\cosh(t) = \cos(it)$, and that $\sinh(t) = -i \sin(it)$, where $i = \sqrt{-1}$.

Thus, $\cosh(1) = \frac{e+e^{-1}}{2}$. We wish to show that this is irrational.

Proposition 5.15. $\cosh(1)$ is irrational.

Proof. We have that

$$\begin{aligned}\cosh(1) &= \frac{e + e^{-1}}{2} = \frac{1}{2} \left(\sum_{n=0}^{\infty} \frac{1}{n!} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \right) = \frac{1}{2} \left(\sum_{n=0}^{\infty} \frac{1 + (-1)^n}{n!} \right) \\ &= \frac{1}{2} \left(\frac{2}{0!} + \frac{2}{2!} + \frac{2}{4!} + \cdots \right) = \frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{(2n)!}.\end{aligned}$$

To show that this is irrational, we shall use our irrationality criterion. Let us choose N such that $\frac{1}{N} < \varepsilon$, and let $q = N!$, $p = \sum_{n=0}^N \frac{N!}{(2n)!}$. Therefore,

$$\begin{aligned}|q \cosh(1) - p| &= \left| N! \sum_{n=0}^{\infty} \frac{1}{(2n)!} - N! \sum_{n=0}^N \frac{1}{(2n)!} \right| = \sum_{n=N+1}^{\infty} \frac{N!}{(2n)!} \\ &= \sum_{n=N+1}^{\infty} \frac{1}{(N+1)(N+2) \cdots (2n)} \leq \sum_{n=N+1}^{\infty} \frac{1}{(N+1)^{2n-N}} = \frac{1}{(N+1)^{N+2}} \cdot \frac{1}{1 - 1/(N+1)^2} \\ &= \frac{1}{(N+1)^{N+2}} \cdot \frac{(N+1)^2}{(N+1)^2 - 1} = \frac{1}{N} \cdot \frac{1}{(N+1)^N} \cdot \frac{1}{N+2} < \frac{1}{N} < \varepsilon.\end{aligned}$$

Thus, by Theorem 3.1, $\cosh(1)$ is irrational. ■

This proof is easily modified to show that $\sinh(1)$ is irrational, and we leave it as an exercise to the reader.

6. TRANSCENDENCE

In this paper, we have looked at irrationality. We will now turn to transcendence.

Definition 6.1. A number α is said to be *algebraic* if it is the root of some nonzero polynomial with integer (or equivalently, rational) coefficients.

Example. The number $i = \sqrt{-1}$ is a root of $x^2 + 1$, and is therefore algebraic.

Example. A rational number $\frac{a}{b}$ is a root of $x - \frac{a}{b}$ (and $bx - a$), and is therefore algebraic.

Example. $\sqrt{2}$ is a root of $x^2 - 2$, and is therefore algebraic.

Remark 6.2. The algebraic numbers form a field (i.e. they are closed under addition, subtraction, multiplication, and division). The simplest proof of this requires Abstract Algebra. We refer the reader to [RSa] to obtain the necessary background. If α and β are algebraic over $\mathbb{Q}$, then we have that $[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}]$ is finite. We shall define this number to be n . Therefore, the numbers $\alpha + \beta, (\alpha + \beta)^2, \dots, (\alpha + \beta)^{n+1}$ are linearly dependent. This gives us a nonzero polynomial with root $\alpha + \beta$. The same is true for $\alpha\beta$ and α^{-1} .

Definition 6.3. A number is said to be *transcendental* if it is not algebraic.

Our main tool will be the Lindemann–Weierstrass theorem.

Theorem 6.4 (The Lindemann–Weierstrass Theorem). *If $u \neq 0$ is algebraic, then e^u is transcendental.*

The proof of this theorem is beyond the scope of this paper. We refer the reader to [Wik20] for a proof.

This theorem will allow us to easily prove many transcendence (and hence, irrationality) results.

Corollary 6.5. *e is transcendental.*

Proof. Take u to be 1. ■

Corollary 6.6. *e^r is transcendental for rational r .*

Proof. Rational numbers are algebraic. Therefore, take $u = r$. ■

Corollary 6.7. *π is transcendental.*

Proof. Let us assume that π is algebraic. Then, we have that $i\pi$ is algebraic. This means that, by Theorem 6.4, $e^{i\pi} = -1$ is transcendental, a contradiction. ■

Corollary 6.8. *π^n is transcendental for nonzero integers n .*

Proof. Let us assume that $n > 0$. Then, if π^n is a root of $p(x)$, π is a root of $p(x^n)$, which is a contradiction. Now, because the algebraic numbers form a field, if π^{-n} is algebraic for $n > 0$, then π^n would be algebraic, which is a contradiction. ■

Corollary 6.9. *$\pi^{a/b}$ is transcendental for rational $\frac{a}{b} \neq 0$.*

Proof. If $\pi^{a/b}$ were algebraic, then $(\pi^{a/b})^b = \pi^a$ would also be algebraic. This is a contradiction. ■

Corollary 6.10. *$\sin(n)$ and $\cos(n)$ are transcendental for nonzero algebraic numbers n .*

Proof. We have that $e^{in} = \cos(n) + i\sin(n)$. Now, we know that $\sin(n) = \sqrt{1 - \cos^2(n)}$. Thus, if $\cos(n)$ were transcendental, then so is $\cos^2(n)$ and $1 - \cos^2(n)$. This means that $\sqrt{1 - \cos^2(n)} = \sin(n)$ is also transcendental. We can also swap $\sin(n)$ and $\cos(n)$ to get that $\sin(n)$ is transcendental if and only if $\cos(n)$ is as well. Now, if $\sin(n)$ is algebraic, so is $\cos(n)$, and therefore, $\cos(n) + i\sin(n)$ is algebraic. Because we are assuming that n is algebraic, this is a contradiction, and so $\sin(n)$ and $\cos(n)$ are transcendental. ■

Corollary 6.11. *$\sin^{-1}(n)$ and $\cos^{-1}(n)$ are transcendental for algebraic nonzero n .*

Proof. Let $\phi = \sin^{-1}(n)$. Then, $\sin(\phi)$ and $\cos(\phi)$ are algebraic. If we assume that ϕ is algebraic, then $e^{i\phi} = \cos(\phi) + i\sin(\phi)$ is transcendental by the Lindemann–Weierstrass Theorem. However, this is an algebraic number, so ϕ must be transcendental. The same is true for $\phi_0 = \cos^{-1}(n)$. ■

Corollary 6.12. *$\cosh(n)$ and $\sinh(n)$ are transcendental for algebraic nonzero n .*

Proof. Let us suppose that n is algebraic. Then, in is also algebraic. Thus, $\sin(in) = i\sinh(n)$ and $\cos(in) = \cosh(n)$ are transcendental. Clearly, if $i\sinh(n)$ is transcendental, so is $\sinh(n)$. ■

Corollary 6.13. *$\ln(n)$ is transcendental for nonzero algebraic n .*

Proof. Let us suppose that $\ln(n)$ is algebraic. Then, we have that $e^{\ln(n)} = n$ is transcendental by Theorem 6.4, which is a contradiction. ■

Another important theorem is the Gelfond–Schneider Theorem.

Theorem 6.14 (The Gelfond–Schneider Theorem). *If $a \neq 0, 1$ is algebraic, and b is an irrational algebraic number, then a^b is transcendental.*

For a proof, we refer the reader to [War12]. This allows us to prove a few more facts.

Corollary 6.15. $\sqrt{2}^{\sqrt{2}}$ is transcendental.

Proof. Take $a = b = \sqrt{2}$ (which we already know is irrational). ■

Corollary 6.16. e^π is transcendental.

Proof. We apply Euler’s Formula to get that

$$e^\pi = e^{-i \cdot i\pi} = (e^{i\pi})^{-i} = (-1)^{-i},$$

so let $a = -1$ and $b = -i$. Therefore, e^π is transcendental. ■

7. OPEN PROBLEMS IN IRRATIONALITY

Despite the fact that almost any person knows what an irrational number is, there are still many questions about which numbers are irrational.

Question 7.1. *Is $e + \pi$ irrational?*

Question 7.2. *Is $e \cdot \pi$ irrational?*

Question 7.3. *Is π^e irrational?*

None of these answers are known, but are all believed to be “yes”. However, there is some work done towards the solutions of the first two questions.

Proposition 7.4. *At least one of $e \cdot \pi$ and $e + \pi$ is irrational.*

Proof. It is clear that both e and π are roots of $(x - e)(x - \pi) = x^2 - (e + \pi)x + e \cdot \pi$. Because both e and π are transcendental, at least one of $e + \pi$ and $e \cdot \pi$ must be irrational. ■

REFERENCES

- [AZHE10] Martin Aigner, Günter M Ziegler, Karl H Hofmann, and Paul Erdos. *Proofs from the Book*. Springer, 2010.
- [Niv61] Ivan Niven. *Numbers: rational and irrational*, volume 1. Random House New York, 1961.
- [Niv85] Ivan Niven. *Irrational numbers*, volume 11. American Mathematical Soc., 1985.
- [Niv00] Ivan Niven. A simple proof that π is irrational. In *Pi: A Source Book*, pages 276–276. Springer, 2000.
- [RSa] Simon Rubinstein-Salzedo. *Abstract Algebra*. Self Published.
- [RSb] Simon Rubinstein-Salzedo. *Infinite Series*. Self Published.
- [War12] Matthijs Warrens. On irrational and transcendental numbers. 2012.
- [Wik20] Wikipedia contributors. Lindemann–weierstrass theorem, 2020. [Online; accessed 30-July-2020].